

Linear Algebra Notes: Vector Spaces

Compiled Notes

Vector Space

Properties of Linearly Dependent and Independent Sets

Definition 1.1: Let V be a vector space and let $S \subseteq V$ be infinite. We say S is linearly dependent if it contains a finite linearly dependent set, otherwise, it is linearly independent.

Theorem 1.1: A set $S = \{x_1, \dots, x_n\} \subseteq V$ is linearly dependent if and only if there exists k such that x_k is a linear combination of the other vectors in S .

Theorem 1.2: A set $S = \{x_1, \dots, x_n\} \subseteq V$ and $x_1 \neq 0$ is linearly dependent if and only if there exists $k > 1$ such that x_k is a linear combination of $x_1, \dots, x_{k-1}$.

Theorem 1.3: Every subset of a finite linearly independent set is linearly independent.

Theorem 1.4: Let V be a vector space over F , and let $S = \{v_1, \dots, v_n\} \subseteq V$ and $T \subseteq \text{ls}(S)$ such that $|T| > |S|$. Then T is linearly dependent.

Properties of Basis and Dimension

Theorem 1.5: Every vector space has a basis.

Theorem 1.6 (Dimension Theorem for Vector Spaces): Any two bases of a vector space have the same cardinality.

Theorem 1.7: Let V be a finite-dimensional vector space with $\dim(V) = n$. If U is a subspace of V , then $\dim(U) \leq \dim(V)$, the equality holds if and only if $\dim(U) = n$.

Theorem 1.8: If a set $S \subseteq V$ is maximal linearly independent, then the linear span of S is the vector space V .

Theorem 1.9: If a subset $S \subseteq V$ is a basis of V , then it is a maximal linearly independent set.

Theorem 1.10 (Basis Deletion Theorem): If $V = \text{ls}(\{\alpha_1, \dots, \alpha_k\})$, then some v_i can be removed to obtain a basis for V .

Theorem 1.11: Let V be a finite-dimensional vector space and let $\dim V = n$. Let $S = \{\alpha_1, \dots, \alpha_n\} \subseteq V$ such that $\text{ls}(S) = V$. Then S is a basis for V .

Theorem 1.12 (Basis Extension Theorem): Every linearly independent set of vectors in a finite-dimensional vector space V can be extended to a basis of V .

Properties of Subspaces

Definition 1.5: A subspace $W \subseteq V$, a vector space over F , must also be a vector space over F with respect to the same operations of addition and scalar multiplication.

Theorem 1.14: $W \subseteq V$ is a subspace of V if and only if for all $x, y \in W$ and $\alpha, \beta \in F$, the vector $\alpha x + \beta y$ is in W .

Theorem 1.15: The union of two subspaces $U, W \subseteq V$ is a subspace of V if and only if $U \subseteq W$ or $W \subseteq U$.

Theorem 1.16: If U, W are subspaces of V , then $U + W$ is the smallest subspace containing both U and W .

Theorem 1.17: If U, W are subspaces of a finite-dimensional vector space V , then $\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$.

Theorem 1.18: $V = U \oplus W$ if and only if every element in V can be uniquely expressed as a sum of an element from U and an element from W .

Inner Product Space

Definition and Example

Definition 2.1: An inner product on a vector space V over a field K is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow K$ that satisfies the following properties:

- Positivity and definiteness: $\langle x, x \rangle \geq 0$ for all $x \in V$, and $\langle x, x \rangle = 0$ if and only if $x = 0$.
- Conjugate symmetry: $\langle x, y \rangle = \overline{\langle y, x \rangle}$ for all $x, y \in V$.
- Linearity in the first argument: $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$ for all $x, y, z \in V$ and $\alpha, \beta \in K$.

Examples include standard inner products on $\mathbb{R}^n$ and $\mathbb{C}^n$, and more complex structures like matrix spaces and function spaces.

Existence of Inner Product

Theorem 2.1: There exists an inner product on every non-trivial vector space over K .

Basic Properties of an Inner Product

Theorem 2.2:

1. $\langle 0, u \rangle = 0$ for every $u \in V$.
2. $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$ for every $u, v, w \in V$.
3. $\langle u, \lambda v \rangle = \lambda \langle u, v \rangle$ for all $u, v \in V$ and $\lambda \in K$.

Orthogonal and Orthonormal Sets

Definition 2.3: Vectors u and v in V are orthogonal if $\langle u, v \rangle = 0$.

Definition 2.4: A set S of vectors in V is called orthonormal if for all $u, v \in S$, $\langle u, v \rangle = \delta_{uv}$ (where δ is the Kronecker delta).

Theorem 2.3: An orthogonal set of non-zero vectors in V is linearly independent.

Orthogonal Complement

Definition 2.5: For $S \subseteq V$, the orthogonal complement of S , denoted $S^\perp$, is the set of all vectors in V that are orthogonal to every vector in S .

Theorem 2.4 (Gram-Schmidt Orthogonalization): Any finite set of linearly independent vectors in V can be transformed into an orthogonal (or orthonormal) set.

Theorem 2.5: Every finite-dimensional inner product space has an orthonormal basis. [non-trivial](#)

Theorem 2.6: For any x in V with orthonormal basis $\{\alpha_i\}$, $x = \sum_{i=1}^n \langle x, \alpha_i \rangle \alpha_i$.

Theorem 2.7: In a finite-dimensional inner product space, the space can be written as the direct sum of any subspace and its orthogonal complement.

Theorem 2.8: The orthogonal complement of the orthogonal complement of a subset S is S itself.

Theorem 2.9: Relationships between subspaces and their orthogonal complements include $(S + T)^\perp = S^\perp \cap T^\perp$ and $(S \cap T)^\perp = S^\perp + T^\perp$.

Theorem 2.10 (Orthogonal Projection Theorem): The orthogonal projection of any vector x onto a subspace W is given by $\sum_{i=1}^k \frac{\langle x, \alpha_i \rangle}{\langle \alpha_i, \alpha_i \rangle} \alpha_i$, where $\{\alpha_i\}$ is an orthonormal basis for W .

Theorem 2.11 (Cauchy-Schwarz Inequality): For any vectors x and y in V , $|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$, with equality if and only if x and y are linearly dependent.

Linear Transformations

Definitions and Examples

Definition 3.1: A mapping $T : V \rightarrow W$ between two vector spaces over the same field F is called a *linear transformation* if it satisfies $T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$ for all $x, y \in V$ and all $\alpha, \beta \in F$.

Example 3.1: Examples of linear transformations include:

- The projection $T : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $T(x, y) = x$.
- Matrix transformations $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T(x) = Ax$ for a matrix $A \in M_{m \times n}(\mathbb{R})$.
- The integration operator $T : C[a, b] \rightarrow \mathbb{R}$ defined by $T(f) = \int_a^b f(x) dx$.

- The zero transformation $T(x) = 0$ for all $x \in V$.
- The identity transformation $T(x) = x$ for all $x \in V$.
- Scalar multiplication $T(x) = \lambda x$ for a scalar $\lambda \in F$.

Theorems on Linear Transformations

Theorem 3.1: Any linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}$ can be expressed as $T(x_1, \dots, x_n) = \sum_{i=1}^n \alpha_i x_i$ where $\alpha_i \in \mathbb{R}$.

Theorem 3.2: For a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, there exist linear transformations $T_i : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $T(x) = (T_1(x), \dots, T_m(x))$ for all $x \in \mathbb{R}^n$.

Theorem 3.3: Given vector spaces V and W over field F , and an ordered basis $\{u_1, \dots, u_n\}$ of V , for any vectors $w_1, \dots, w_n$ in W , there is a unique linear transformation $T : V \rightarrow W$ such that $T(u_j) = w_j$ for $j = 1, \dots, n$.

Theorem 3.4: Extends Theorem 3.3 to infinite dimensional spaces, establishing a unique linear transformation $T : V \rightarrow W$ given a basis $\{u_\alpha : \alpha \in I\}$ of V and vectors $\{w_\alpha : \alpha \in I\}$ in W .

Properties of Linear Transformations

Definition 3.2:

- $\text{Ker}(T)$: $\{x \in V : T(x) = 0\}$ is called the kernel of T , a subspace of V .
- $R(T)$: $\{T(x) : x \in V\}$ is called the range of T , a subspace of W .

Definition 3.3: The dimensions of the range and kernel of T are called the rank and nullity of T , respectively.

Theorem 3.5: A linear transformation $T : V \rightarrow V$ is one-to-one if and only if the kernel of T is $\{0\}$.

Theorem 3.6: If $T(u_1), \dots, T(u_n)$ are linearly independent in W , then $u_1, \dots, u_n$ are linearly independent in V .

Theorem 3.7: If T is one-to-one and $u_1, \dots, u_n$ are linearly independent in V , then $T(u_1), \dots, T(u_n)$ are linearly independent in W .

Theorem 3.8: If B is a basis of V , then the range of T is contained within the linear span of $T(B)$.

Theorem 3.9: (Rank-Nullity Theorem) For $T : V \rightarrow W$, $\text{Nullity}(T) + \text{Rank}(T) = \dim(V)$.

Theorem 3.10: Relations between dimensions of vector spaces and properties of linear transformations, including injectivity and surjectivity criteria based on dimensions.

Eigenvalues and Eigenvectors

Definitions

Definition 4.1: A scalar λ is said to be an eigenvalue of a matrix A if there exists a non-null vector x such that $Ax = \lambda x$. Any such x is called an eigenvector of A corresponding to the eigenvalue λ .

Definition 4.2: The polynomial $\det(xI - A)$ is called the characteristic polynomial of A , and the equation $\det(xI - A) = 0$ is called the characteristic equation.

Definition 4.3: An eigenpair of a matrix A is a pair (λ, x) , where x is an eigenvector of A corresponding to the eigenvalue λ .

Definition 4.4: The algebraic multiplicity of an eigenvalue λ is the number of times λ appears as a root of the characteristic polynomial of A .

Definition 4.5: The eigenspace $E_\lambda(A)$ associated with an eigenvalue λ is the set $\{x \in F^n : (A - \lambda I)x = 0\}$. The dimension of E_λ is called the geometric multiplicity of λ .

Theorems

Theorem 4.1: A scalar λ is an eigenvalue of a matrix A if and only if λ is a root of the polynomial $\det(xI - A)$.

Theorem 4.2: Every matrix A in $M_n(\mathbb{C})$ has at least one eigenvalue.

Theorem 4.3: Similar matrices have the same characteristic polynomial, but the converse is not true.

Theorem 4.4: For a block matrix $C = \begin{bmatrix} A & B \\ 0 & D \end{bmatrix}$, the characteristic polynomial of C , $P_C(x)$, is the product of the characteristic polynomials of A and D , i.e., $P_C(x) = P_A(x)P_D(x)$.

Theorem 4.5: If $f(x)$ is a polynomial and (λ, x) is an eigenpair of matrix A , then $(f(\lambda), x)$ is an eigenpair of $f(A)$.

Theorem 4.6 to 4.16: Various properties and structural insights into matrices concerning their eigenvalues and eigenvectors, including block matrices, polynomial transformations, and relationships between algebraic and geometric multiplicities.

Theorem 4.17: The space of all matrices $M_n(F)$ is closed under addition and multiplication. Specifically, if A, B are in $M_n(F)$, then $A + B$ and AB are also in $M_n(F)$.

Theorem 4.18: If A is a matrix representation of a linear transformation T in some basis, and λ is an eigenvalue of T , then λ is also an eigenvalue of A .

Theorem 4.19: The minimal polynomial $m_A(x)$ of a matrix A divides any polynomial $p(x)$ such that $p(A) = 0$.

Theorem 4.20: A matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces equals the dimension of the space F^n , and all its eigenvalues' algebraic and geometric multiplicities match.

Cayley-hamilton theorem

Theorem 4.21: The eigenvalues of a matrix A sum to its trace, and their product equals the determinant of A .

Theorem 4.22: A matrix is diagonalizable if and only if the minimal polynomial has distinct roots.

Theorem 4.23: Hermitian matrices have real eigenvalues and orthogonal eigenvectors corresponding to distinct eigenvalues.

Types of Operators

Adjoint and Their Properties

Definition 5.1: Let T be a linear operator on an inner product space V . Then T has an adjoint on V if there exists a linear operator T^* on V such that $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ for all x and y in V .

Theorem 5.1: Let $(V, \langle, \rangle)$ be a finite dimensional inner product space over K and let f be a linear functional. There exists a unique vector y in V such that $f(x) = \langle x, y \rangle$ for all x in V .

Theorem 5.2: For any linear operator T on a finite-dimensional inner product space $(V, \langle, \rangle)$, there exists a unique linear operator T^* on V such that $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ for all x, y in V .

Theorem 5.3: Let $(V, \langle, \rangle)$ be a finite dimensional inner product space over K and let $B = \{u_1, \dots, u_n\}$ be an orthonormal basis for V . Let T be a linear operator on V and let A be the matrix of T in the ordered basis B . Then $A_{kj} = \langle T(u_j), u_k \rangle$.

Theorem 5.4: Let $(V, \langle, \rangle)$ be a finite dimensional inner product space over K and let T be a linear operator on V . In any orthonormal basis for V , the matrix of T^* is the conjugate transpose of the matrix of T .

Special Types of Operators

Definition 5.2: A linear operator T is called a self-adjoint operator (or Hermitian operator) if $T = T^*$.

Definition 5.3: A linear operator T is called a Skew-Hermitian operator if $T = -T^*$.

Definition 5.4: A linear operator T is called a Unitary operator if $TT^* = T^*T = I$.

Definition 5.5: A linear operator T is called a Normal operator if $TT^* = T^*T$.

Remark 5.1: Let $(V, \langle, \rangle)$ be a finite dimensional inner product space over K and let T be a linear operator on V .

1. T is self-adjoint if $[T]_B$ is Hermitian for any orthonormal basis B .
2. T is Skew-Hermitian if $[T]_B$ is Skew-Hermitian for any orthonormal basis B .
3. T is Unitary if $[T]_B$ is Unitary for any orthonormal basis B .

4. T is normal if $[T]_B$ is Normal for any orthonormal basis B .